

Exam II Review

Section	Topics	Exam Prototypes
1.3	Definition of a cyclic group Definition of a generator Theorem 1.3.11 Theorem 1.3.23 <i>Every subgroup of a cyclic group is cyclic</i> Theorem 1.3.25 Subgroup lattices	Definitions 1.3.4 Problems, Page 60 1*, 4*, 6*, 7*, 11*, 12*, 17*, 19, 24
1.4	Definition of a permutation Composition of permutations $(S_n, \circ)$ Two line form of a permutation in S_n Definition of a cycle One line form a permutation in S_n Theorem 1.4.20 <i>Every permutation in S_n can be written as a product of disjoint cycles</i> Proposition 1.4.22 Lemman 1.4.26 <i>Every permutation can be written as a product of 2-cycles</i> Definitions 1.4.32 (Even/odd permutations) Theorem 1.4.34 A_n is a subgroup of S_n Theorem 1.4.36 $ A_n = \frac{ S_n }{2} = \frac{n!}{2}$	Definitions 1.4.6, 1.4.11, 1.4.32 Problems, Page 71 6*, 7*, 10*, 11*, 12*, 13*, 18, 28, 30
2.1	Theorem 2.1.3 Definition 2.1.4 Lemma 2.1.8 Theorem* 2.1.9 <i>Lagrange's Theorem</i> Definition 2.1.10 Theorem 2.1.16 Theorem 2.1.18 Theorem 2.1.19	Definitions 2.1.4, 2.1.10 Problems, Page 78 3*, 5, 6, 8, 9a*, 14*, 15*, 18, 20*, 25*